

OM CHAUHAN

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EDUCATION

Master of Science in Software Engineering

Arizona State University, Tempe, AZ

Aug 2024 – May 2026

GPA: 3.78/4.0

Bachelor of Engineering in Computer Engineering

Gujarat Technological University, India

Aug 2020 – May 2024

GPA: 3.39/4.0

EXPERIENCE

Software Engineer Intern

Axisray, Ahmedabad, India

June 2023 – Jun 2024

- Engineered a fullstack HR automation system using **Spring Boot** and **MySQL**, reducing manual HR operations for a **50+ person** team
- Built a voice-driven HR query interface by integrating **Amazon Alexa Skills Kit** with **AWS Lambda**, enabling employees to retrieve HR information hands-free through natural language requests
- Automated the build and deployment pipeline using **AWS CodePipeline**, **CodeBuild**, **CodeDeploy**, and **CodeArtifact** — triggered via **GitHub** webhooks, reducing deployment errors to near-zero
- Developed and trained a real-time fire detection system from scratch using **YOLOv5** on a custom dataset of **10,000 images and 100 videos**, achieving **95% accuracy** on a held-out test set with a **40% reduction** in false positive alerts

PROJECTS

SafeGuard | Mobile SOS Application

- Architected and migrated AWS infrastructure from **ECS Fargate + ALB + ElastiCache** to a single **EC2** instance with **Docker Compose**, cutting projected monthly cloud spend from **\$50 to \$4/month (92% reduction)** to meet zero-budget constraints
- Implemented multi-phase location tracking logic in **Node.js** and **Socket.io** handling four operational states — active SOS trigger, passive background watching, wake-on-push for killed app processes, and an offline dead-zone protocol — enabling reliable coordination across degraded network conditions
- Authored modular **Terraform** infrastructure (**VPC**, **ECS**, **ElastiCache Redis**, **Cognito**, **Secrets Manager**) with **GitHub Actions CI/CD** using **OIDC authentication**, eliminating long-lived AWS credentials

StratusVault | Secure Cloud File Share on GCP

- Designed and implemented a cloud-native document storage service on **GCP** using **Cloud Run** for the stateless **Spring Boot API**, **Cloud SQL (PostgreSQL)** for relational metadata, and **Cloud Storage** for private object storage
- Built a file-level ACL system using **Cloud SQL** and **Firebase Authentication** — verifying Firebase ID tokens server-side, resolving per-file owner and shared-user permissions from the database, with no public **GCS** bucket exposure

Hybrid Cloud & Edge AI System

- Built a queue-depth autoscaling controller that dynamically provisioned and terminated up to 15 EC2 app-tier instances based on SQS backlog, scaling from zero under idle conditions and achieving sub **1.2s average latency** across a 100-request benchmark
- Architected an edge computing pipeline using **AWS IoT Greengrass** and **MQTT** to run **MTCNN** face detection locally on an **EC2 IoT** client — transmitting only detected face crops (not raw frames) to **SQS** for cloud-side **FaceNet** recognition, reducing cloud processing overhead
- Deployed a serverless **AWS Lambda** inference backend with **SQS queues**, **S3 persistence**, and containerized **PyTorch** models — achieving 100% benchmark accuracy

TECHNICAL SKILLS

Languages: Python, Java, JavaScript/TypeScript, C/C++, SQL

Backend & Systems: Node.js, Spring Boot, REST APIs, WebSockets, Socket.io, Django, Flask, Git

Cloud & Infrastructure: AWS (ECS Fargate, EC2, Lambda, S3, RDS, ElastiCache, ALB, Cognito, Secrets Manager, SNS, IoT Greengrass), GCP (Cloud Run, Cloud SQL, Cloud Storage), Terraform, GitHub Actions, Docker

Databases: Redis, MongoDB, MySQL, PostgreSQL, GraphQL